

4. Model Chapter

4.1. The problem the model is built to explain

Our model begins with a simple organizational reality: individuals enter with measurable characteristics and prior performance, develop alongside peers in a shared environment, are evaluated over time, and eventually face decisions that distinguish some from others. Those decisions, such as promotion, selection, admission, drafting, tenure, or other scarce opportunities, are observable signals of what the organization defines as success. The central problem of this chapter is to explain how an individual's environment can shape the probability of that later success. Individuals do not develop or compete in isolation. They are assigned to groups that include peers, their performance is evaluated within those environments, and only afterward are some selected for a limited opportunity. These processes are related, but they are not the same process. We therefore represent advancement as three distinct stages: ASSIGN, SCORE, and SELECT. In this way, the mechanisms that generate peer environments can be separated from the mechanisms that translate performance into outcomes.

Our model does not seek to reproduce every institutional detail of the empirical domains we study. Instead, it provides a minimal framework for asking when group composition changes individual competitive outcomes, which mechanisms are necessary for that effect to emerge, and how the resulting patterns depend on the structures of assignment, scoring, and selection.

4.1.1. Why strong environments do not always produce greater success.

At first glance, stronger environments appear advantageous. High-performing groups often contain individuals who later achieve desirable outcomes at higher rates, making it natural to associate group quality with individual success. Our empirical results, however, suggest that this relationship is not uniformly positive across all individuals or all parts of the performance distribution. In some settings, the apparent advantage associated with stronger peers weakens among individuals near the upper end of the distribution and, under some conditions, can reverse.

This pattern presents the motivating puzzle for the model. If stronger groups generally provide access to better peers, information, norms, resources, or developmental opportunities, why should the relationship between peer quality and later success ever weaken? Answering that question requires us to separate the value of the environment from the process through which later outcomes are awarded.

4.1.2. Separating environmental effects from later selection.

The quality of an individual's environment and the probability of later selection describe different parts of the organizational process. A strong environment may improve learning, development, information, visibility, or performance. The same environment may also introduce competitive challenges that ascribe relative standing or create difficulty distinguishing oneself from nearby competitors. These effects occur before the organization converts individual performance into a later outcome.

Selection therefore cannot be treated as a direct measure of environmental value. An individual may benefit from a strong environment without ultimately being selected, while another may receive a favorable outcome for reasons that depend on the rules governing

evaluation and selection rather than on environmental benefits alone. The model separates these processes so that the effects associated with the peer environment are not automatically conflated with the institutional process that produces success.

4.1.3. A simple organizational example.

Consider two individuals who enter an organization with similar previously measured performance but are assigned to different groups. One develops in a group whose members are mostly moderate performers. The other develops among substantially stronger peers. The second individual may benefit from the knowledge, expectations, information, or other advantages associated with the stronger environment. At the same time, that individual may occupy a weaker relative position within the group or face greater difficulty distinguishing himself or herself from nearby competitors.

Suppose both individuals are later considered for the same type of organizational opportunity. Their initial characteristics may be similar, but the environments in which they developed, the way their performance is evaluated, and the rule used to allocate the opportunity may differ. The peer environment is often local, while selection may occur locally or across a broader global organizational population. In some settings the same group substantially defines both processes; in others, individuals who developed in many different groups later compete against one another. Understanding the final outcome therefore requires more than knowing whether one individual belonged to a stronger group.

4.1.4. The three-stage architecture: ASSIGN, SCORE, and SELECT.

We represent this process using three stages: ASSIGN, SCORE, and SELECT. ASSIGN determines how individuals are placed into groups and therefore which peer environments they experience. SCORE describes how individual performance and characteristics of the surrounding environment combine to determine competitive standing. SELECT specifies how those scores are converted into observable outcomes.

Separating the process into these stages allows each mechanism to be examined independently. Changes in assignment can alter peer environments without changing the scoring rule. Environmental effects can change scores without changing who is ultimately selected. Likewise, the same set of scores can generate different outcomes under different selection rules or levels of opportunity. The model therefore provides a framework for identifying which stage, or interaction among stages, is responsible for the patterns observed in the empirical data.

4.2. Individuals, groups, and peer environments

4.2.1. The individual entering the model.

The model begins with an individual-level quantity A_i that represents the performance or characteristic used to describe person i in the modeled process. Depending on the empirical setting, A_i may be measured before assignment or may represent performance observed after individuals have already spent time in an environment. The model does not require A_i to necessarily represent innate ability or a context-independent trait. Rather, A_i is treated as fixed within a given model experiment so that the effects of assignment, scoring, and selection can be examined separately. In settings where the environment itself may alter observed performance, treating A_i as fixed is a simplifying assumption that separates the

assignment and selection mechanisms from the process through which performance develops over time.

4.2.2. The group as the individual's environment.

Individuals are embedded in groups whose composition defines the immediate peer environment considered by the model. These organizational structures can be represented as bipartite networks in which one node set represents individuals and the other represents groups, with membership encoded as edges between the two sets. The peer relationships among individuals are then induced by shared group membership and can be represented through the individual projection of the bipartite network. A group may be a military unit, athletic team, academic department, school, or another organizational setting in which individuals develop and are evaluated alongside others. Group membership determines which peers are locally relevant to person i and therefore which observations contribute to the environmental quantities associated with that individual.

The model also requires the group to have a defined size or capacity. Capacity matters because assignment is not unconstrained: individuals cannot all be placed into the same group, and the number of available positions shapes the set of possible assignments. Group size therefore affects both the composition of peer environments and the assignment process that generates them. The group is treated as the local environment in which peer exposure occurs, even when later selection may take place across a broader organizational population. In network terms, individuals and groups form an affiliation structure: membership links individuals to groups, and shared group membership induces the local peer network through which individuals experience one another.

4.2.3. Measuring peer quality.

To characterize the environment surrounding individual i , we summarize the measured characteristics of the individual's peers. A natural measure is the mean performance or characteristic of the other members of the group. When constructing this quantity, the focal individual's own value is excluded so that the environmental measure reflects the peers surrounding the individual rather than mechanically incorporating the individual's own performance.

Let the peer-quality measure for individual i summarize the values of the other members of i 's group. Higher values indicate that the focal individual is surrounded, on average, by stronger peers according to the characteristic represented by A . This quantity provides a common way to compare peer environments across individuals and groups, even when the underlying empirical measure differs across domains.

The peer-quality measure is descriptive. It tells us what kind of environment the individual occupies. It does not, by itself, specify whether that environment helps or harms the individual.

4.2.4. From peer quality to congestion.

Strong peer environments may provide benefits, but they may also impose competitive costs. We use the term congestion to describe the adverse competitive effects that arise when an individual is surrounded by other strong performers. Congestion therefore represents a

modeled drawback of the peer environment rather than a neutral description of that environment.

Peer quality and congestion are related but distinct. Peer quality describes the composition of the individual's surrounding group, while congestion describes one possible consequence of that composition for competitive standing. A group may contain strong peers and therefore have high peer quality, but whether those peers impose a substantial penalty depends on the scoring mechanism specified in the model. Mean peer performance is therefore an environmental descriptor; congestion is a modeled negative effect associated with that environment.

This distinction is important because strong peers may simultaneously provide developmental benefits and create competitive disadvantages. Later, we describe this more generally as a balance between environmental benefits and drawbacks. In the present model, we simplify that broader relationship by focusing on the drawback side and representing its adverse competitive effect through congestion.

4.2.5. A concrete example concrete example from the Army

The Army provides a straightforward example of the distinction between the individual, the local peer environment, and later organizational selection. An officer arrives at a unit with an existing performance history and is assigned to a local group of peers. The composition of that group defines the immediate environment in which the officer serves, is evaluated, and continues to accumulate performance information.

Peer quality can therefore be described using the characteristics of the other officers in the local environment, excluding the focal officer's own value. A stronger local group may provide developmental benefits, but it may also make relative distinction more difficult. The peer environment describes the setting in which the officer develops and is evaluated; it does not itself determine the later outcome.

Promotion provides the subsequent selection process. The decision is not made by the officer's immediate unit but across a broader Army population. The model therefore separates the local environment in which performance develops from the broader global selection process that converts accumulated performance into advancement.

Core notation used throughout the model

- A_i — measured characteristic/performance of individual i
- $g(i)$ — group containing individual i
- T_j — evolving, then final, centroid of group j
- $L_{Q,i}$ — leave-one-out peer-quality measure around individual i
- L_C — conceptual congestion mechanism
- C_j — implemented group-level congestion quantity
- S_i — competitive score presented to SELECT

4.3. ASSIGN: generating peer networks from bipartite organizational structure

The ASSIGN stage determines the organizational structure within which peer environments arise. From a network perspective, this is a constrained affiliation problem: individuals are assigned to groups, group capacities limit the number of available memberships, and the resulting affiliations determine who is exposed to whom. Shared group membership therefore creates the local peer neighborhoods used in the later stages of the model.

The LG assignment mechanism was developed from this network-science perspective. Like a configuration-style model, it preserves structural constraints on group membership while allowing the pattern of who is grouped with whom to vary. Unlike purely random assignment, however, the mechanism can introduce a preference for similarity between an arriving individual and the members already assigned to a group. This allows the model to vary assortative group formation independently of the subsequent SCORE and SELECT mechanisms.

4.3.1. Organizational environments as bipartite networks

The organizational settings considered in this dissertation can be represented naturally as bipartite networks. One set of nodes represents individuals, while the second represents the organizational groups to which they belong, such as military units, athletic teams, academic departments, or schools. Edges connect individuals only to groups, not directly to other individuals within the bipartite representation.

This representation separates the organizational membership structure from the peer relationships induced by that structure. Individuals become peers because they share membership in the same group. The peer network among individuals therefore appears in a one-mode projection of the bipartite network, in which two individuals are connected when they share a common group. The bipartite network captures assignment; its projection reveals the local peer structure produced by that assignment.

4.3.2. The bipartite configuration model as a null model of assignment

Network science commonly evaluates empirical structure by comparing an observed network with an appropriate randomized reference network. The purpose of the reference model is not to remove all structure, but to preserve selected empirical constraints while randomizing the remaining degrees of freedom. The configuration model provides a canonical example: a network is randomized while preserving a specified degree sequence.

The same logic applies to a bipartite organizational network. From the empirical system we know the two sets of nodes, the number of individuals and groups, and the observed degree sequence on each side. Individual-node degree records the number of group memberships represented in the network, while group-node degree records group size or capacity. These degrees define the number of edge stubs associated with each node. A bipartite configuration model generates randomized networks by reconnecting these stubs subject to the restriction that edges connect nodes from opposite layers.

This produces a structural null model that preserves the empirical opportunity structure while randomizing who belongs to which group.

4.3.3. Introducing homophily into the bipartite configuration model

The organizational groups we study are not well represented by a model of purely random assignment. The ordinary bipartite configuration model provides a randomized assignment

baseline, but it does not capture the possibility that assignment may depend systematically on individual characteristics. In the domains considered here, recruiting, self-selection, screening, specialization, and institutional sorting can make some individuals more likely than others to enter particular groups. When these processes favor similarity on the characteristic of interest, they can produce homophilic assignment and, ultimately, assortative group composition.

The model begins with two sets of nodes: individual nodes and group nodes. During the modeled period, each individual belongs to exactly one group, so every individual node has degree

$$k_i = 1.$$

The degree of each group node is determined by its observed size or capacity. The complete bipartite degree sequence is therefore specified in advance: each individual contributes one edge stub, while each group contributes a number of stubs equal to the number of individuals it must contain. The network-generation problem is to connect every individual stub to a group stub while preserving these empirical degree constraints.

To introduce homophily, each individual carries the measured characteristic A_i , and each group j is characterized during assignment by an evolving centroid T_j . Before any individuals are assigned, all groups begin with the common reference value

$$T_j = \bar{A}$$

where $\bar{A}$ is the mean of the individual characteristic in the population being assigned.

Individuals are then considered sequentially in randomized order. At each step, the focal individual can connect only to groups that retain unmatched stubs. Among those eligible groups, assignment occurs preferentially toward groups whose current centroid T_j is more similar to the focal individual's A_i . We introduce the parameter ρ to control the strength of this similarity preference.

When the first individual joins a group, that individual's A_i becomes the group's centroid T_j . Thereafter, T_j is updated after every assignment to equal the mean A_i of the group's current members. Thus, group characteristics are not fixed in advance. They emerge endogenously as the network is formed. Once assignment is complete, T_j is the final emergent centroid of group j , and every individual and group node has received the number of edges specified by the empirical degree sequence.

This construction separates two properties of the network. The degree sequence constrains how many edges each node ultimately receives, while the homophilic assignment mechanism influences which allowable edges are preferentially formed. The model therefore preserves the empirical structural constraints of the bipartite network while allowing assortative group composition to emerge from similarity in individual-node metadata.

When $\rho = 0$, similarity does not influence assignment, and the model reduces to the capacity-constrained random reference. As ρ increases, assignment becomes increasingly biased toward groups whose evolving centroids resemble the focal individual's A_i . Importantly, ρ describes the strength of the generative preference during network formation; it does not measure the assortativity realized in the completed network. That realized structure must be measured separately after assignment is complete.

4.3.4. From bipartite assignment to projected peer networks

The output of ASSIGN is a bipartite individual-group network. The peer environment experienced by individuals can be represented through the corresponding individual projection, in which two individuals are connected when they share membership in the same organizational group. The projection therefore makes explicit the peer relationships induced by the assignment process.

This projected network is not a separate object that the present model seeks to analyze in full. Rather, it provides the network representation of the local peer environment generated by ASSIGN. Because those peer ties arise from the underlying bipartite memberships, preserving the empirical group-size and membership constraints during assignment is important for generating realistic local peer neighborhoods. In this way, the bipartite network describes who belongs where, while the projected network describes who is locally exposed to whom. The latter provides the peer structure that is carried forward into the SCORE stage.

4.3.5. Generative preferences versus realized sorting

The similarity preference used to generate the network and the amount of assortative structure realized in the completed network are not the same quantity. The parameter ρ governs how strongly similarity influences assignment during network formation, but because assignment is stochastic and capacity constrained, a given value of ρ does not guarantee a particular level of sorting in the resulting groups.

We therefore require a separate statistic to quantify the degree of sorting that actually emerges after assignment is complete. We denote this quantity H_{sort} . Conceptually, H_{sort} is an ANOVA-style measure of the proportion of total variation in individual A_i that is explained by group membership. If individuals within the same group have similar values of A_i , within-group variation is small relative to variation in the population as a whole and H_{sort} is high. If individuals with different A_i values are distributed more uniformly across groups, group membership explains less of the total variation and H_{sort} is lower.

$$H_{\text{sort}} = 1 - \frac{\sum_i (A_i - \mu_{g(i)})^2}{\sum_i (A_i - \bar{A})^2},$$

with:

- A_i : individual performance characteristic;
- $g(i)$: group containing individual i ;
- $\mu_{g(i)}$: Mean A of that group;
- $\bar{A}$: population mean.

Thus, whereas ρ is an input to the generative process, H_{sort} is an observed property of the network that process produces. Realized sorting depends not only on ρ , but also on the distribution of A_i , the degree sequence, group capacities, randomized arrival order, and the stochastic history of previous assignments. Two networks generated with the same value of ρ may therefore produce different values of H_{sort} .

Finite groups can exhibit nonzero sorting even under random assignment. H_{sort} must therefore be interpreted relative to random reference ensemble generated under the same structural constraints. The relevant question is not simply whether H_{sort} differs from zero,

but whether the realized level of sorting differs from what would be expected under capacity-constrained random assignment.

The purpose of ASSIGN is therefore not to reproduce every topological property of the empirical network. It is to generate controlled peer environments that preserve the observed structural constraints while allowing the composition of those environments to vary systematically. With the peer environments established, the model next asks how membership in those environments changes an individual's competitive standing.

4.4. SCORE: translating peer environment into competitive standing

4.4.1. From peer environment to individual score

The ASSIGN phase determines who is grouped with whom and therefore establishes the peer environment each individual experiences. Once that structure is formed, however, ASSIGN has completed its role. It does not specify whether the resulting environment benefits or disadvantages the individual, nor does it determine how those environmental effects enter later competition.

SCORE introduces that second question. The individual enters the scoring stage with a measured characteristic or performance quantity A_i , while the group generated by ASSIGN provides the surrounding peer environment. SCORE specifies how characteristics of that environment modify the individual's competitive standing relative to others. In this way, the model separates the process that creates peer exposure from the mechanism through which that exposure may affect later outcomes.

This distinction is important because the same assignment structure can support different scoring mechanisms. A group of strong peers might improve development, impose competitive costs, or produce both effects simultaneously. ASSIGN determines the environment; SCORE determines how that environment is translated into the quantity that will later be presented to SELECT.

4.4.2. Environmental benefits, drawbacks, and the modeling simplification

Conceptually, we denote the net effect of the peer environment on the individual's development or competitive position as

$$L = B - D$$

where B represents the aggregate benefits associated with the environment and D represents its aggregate drawbacks. This expression provides a general conceptual account of environmental influence; the present model does not attempt to model both components separately. Instead, it isolates one hypothesized drawback mechanism, competitive congestion, in order to examine whether that mechanism is sufficient to alter later outcomes.

4.4.3. Defining congestion

Peer quality alone does not specify the competitive drawback imposed by a group. Consider the average performance of the peers surrounding individual i . We denote this descriptive quantity $L_{Q,i}$, the leave-one-out mean of A among the *other* members of the group $g(i)$. $L_{Q,i}$ answers a straightforward question: how strong, on average, are the peers surrounding the focal individual?

Congestion asks a different question. The competitive difficulty created by a group may depend less on average peer performance than on the concentration of peers who are sufficiently strong to compete for the same distinction. We use L_C to refer generally to this congestion mechanism. In the implemented model, congestion is operationalized as a group-level quantity C_j , which summarizes the competitive pressure generated by the members of group j . Whereas $L_{Q,i}$ describes the quality of the peer environment surrounding individual i , C_j represents the modeled competitive drawback associated with membership in group j .

That distinction is necessary because groups with similar average peer quality need not generate the same competitive pressure. The competitive consequences of the peer environment are likely to be most important near the **selection margin**, the region of the performance distribution in which relatively small changes in competitive standing could alter eventual selection. Individuals far above or far below that boundary may be less sensitive to modest changes in the surrounding competitive environment, while those near the margin can be disproportionately affected.

A group containing several very strong individuals and several weaker individuals may therefore have the same average peer quality as a more homogeneous group while creating a substantially different competitive environment. Congestion is intended to capture this difference. Rather than depending only on the mean value of A_i within the group, the implemented congestion quantity C_j depends on how the performance of group members is distributed relative to a relevant performance threshold.

The model represents this idea by allowing each member of group j to contribute to C_j according to that individual's value of A_i . Individuals well below the relevant threshold contribute little to congestion, while individuals near or above the threshold contribute more. These individual contributions are aggregated to form the group-level congestion quantity C_j , which is then carried into SCORE as the modeled competitive penalty associated with membership in group j .

We return to the selection margin explicitly in SELECT, where the number of available opportunities determines whether differences in competitive score ultimately change who is selected.

4.4.4. The smooth contribution function

The conceptual congestion mechanism L_C represents the competitive pressure created by sufficiently strong peers within a group. To operationalize that idea, the model must determine how much each group member contributes to the group's overall congestion. Not every peer should contribute equally: an individual far below the relevant competitive threshold should add little to the modeled pressure, while a stronger individual should contribute more.

We therefore map each individual characteristic A_i into a smooth congestion contribution using the logistic function:

$$\sigma[\gamma(A_i - \theta)] = \frac{1}{1 + \exp[-\gamma(A_i - \theta)]}$$

This quantity is not itself the full group-level congestion term. Rather, it is the contribution of individual i to the congestion mechanism. These individual contributions are aggregated across the members of a group in the next step to produce the operational group-level congestion quantity C_j .

We introduce the parameter θ to identify the performance level around which an individual becomes competitively relevant to the congestion mechanism. Individuals with A_i well below θ contribute little, while those well above θ contribute substantially more. The parameter γ controls how sharply this transition occurs. Small values of γ produce a gradual transition, whereas larger values increasingly approximate a hard threshold separating low-contributing from high-contributing peers.

The smooth form avoids treating two individuals with nearly identical performance as categorically different merely because they fall on opposite sides of a cutoff. At the same time, it preserves the substantive intuition underlying congestion: stronger peers, particularly those near or above the relevant competitive threshold, contribute more to the group's competitive pressure.

4.4.5. From group congestion to individual score

The individual contributions defined above are next aggregated at the group level to form C_j , the operational group-level realization of the conceptual congestion mechanism L_C . For group j containing n_j individuals, we define congestion as

$$C_j = \frac{1}{n_j} \sum_{h \in j} \sigma[\gamma(A_h - \theta)]$$

Thus, C_j measures the average concentration of competitively relevant individuals in group j . A group containing relatively few individuals near or above the threshold will have low congestion, while a group containing many such individuals will have high congestion.

In the current specification, C_j is a group-level quantity. Every individual assigned to group j therefore experiences the same modeled congestion, regardless of that individual's own value of A_i . If $g(i)$ denotes the group containing individual i , the competitive score is

$$S_i = A_i - \lambda C_{g(i)}$$

where λ controls the strength with which group congestion reduces competitive standing.

This equation completes the SCORE mechanism. A_i represents the individual performance quantity entering the stage, while $C_{g(i)}$ represents the modeled competitive drawback associated with the individual's group environment. When $\lambda = 0$, the environment has no effect on competitive score and $S_i = A_i$. As λ increases, differences in group congestion have greater influence on the ordering of individuals across groups.

Because C_j is common to every member of group j , this specification has an important consequence for ranking: it can alter comparisons between individuals in different groups without changing their ordering within the same group. We examine that property directly in the next subsection.

4.4.6. What the SCORE mechanism changes, and what it does not

Because congestion is defined at the group level, every member of group j receives the same congestion adjustment. SCORE therefore preserves all pairwise rankings among members of the same group. If two individuals i and h belong to the same group and $A_i > A_h$, then subtracting the same quantity λC_j from both preserves that ordering:

$$S_i - S_h = A_i - A_h$$

Congestion can, however, change comparisons between individuals who belong to different groups. Two individuals with similar values of A_i may receive different competitive scores if their groups generate different levels of congestion. An individual with greater measured performance can therefore fall below an individual with lower measured performance if the difference in group-level congestion is sufficiently large.

This distinction is central to the mechanism. SCORE does not directly reorder peers within a local group; instead, it changes how individuals emerging from different peer environments compare when they later enter a broader selection population. In this sense, the local environment modifies competitive standing across groups while preserving the performance ordering within each group.

The presence of a congestion penalty also does not imply that selection outcomes must change. SCORE changes the quantities presented to SELECT, but only score changes near the eventual selection margin can alter who receives a limited opportunity. Congestion may therefore be active in the scoring process without changing the identities of selected individuals under a particular selection rule or level of scarcity.

4.4.7. Model variants and historical choices

The model developed through several related congestion specifications. These variants are useful for understanding how the current formulation was chosen, but they should be distinguished from the canonical specification described above.

- 4.4.7.1. **Leave-one-out versus full-group congestion** - Earlier implementations defined congestion from the peers surrounding each focal individual, excluding the focal individual's own contribution. This produced an individual-specific leave-one-out congestion quantity. The current specification instead defines congestion over the full group, including every eligible member, and assigns the resulting C_j uniformly to all members of group j . This makes congestion a property of the shared environment rather than a different environmental measure for each focal individual. It also aligns the congestion term with the model's interpretation of the group as a common local environment experienced by all member and produces the within-group ranking property described above.
- 4.4.7.2. **Raw versus scaled congestion** - A second modeling choice concerns scale. The smooth congestion quantity C_j is naturally bounded, while A_i is often standardized. Applying the two quantities directly in the same score equation can therefore make the practical influence of λ depend heavily on their relative numerical scales. Some implementations consequently rescale congestion using a fixed reference distribution before applying λ . This transformation changes the magnitude of the congestion term but not the underlying group composition or the fact that all members of a group receive the same group-level value.
- 4.4.7.3. **Fixed versus selection-related viability thresholds** - The location of the viability threshold θ has also varied across model versions. In some experiments, θ is fixed independently of the later selection rule and identifies the performance region in which peers begin to contribute meaningfully to congestion. Other exploratory formulations linked the threshold more closely to the performance distribution associated with eventual selection. These choices reflect different substantive interpretations of who constitutes a competitively relevant peer and should not be treated as interchangeable.

Across these variants, the central SCORE mechanism remains unchanged: group composition is translated into a congestion quantity, congestion modifies competitive score, and selection remains a separate subsequent process. The canonical specification used in the remainder of the chapter therefore provides a common baseline while the historical variants clarify which results depend on particular modeling choices.

4.5. **SELECT: converting competitive standing into outcomes**

The SCORE stage produces a competitive score S_i for every individual, but a score alone is not an outcome. SELECT specifies the institutional rule that converts the distribution of scores into observed distinctions such as promotion, drafting, tenure, or admission. This separation is essential because the same set of scores can generate different outcomes under different levels of scarcity or different selection rules.

Selection therefore introduces a new feature of the model: capacity. Whereas ASSIGN constrains the number of individuals who can occupy each local group, SELECT constrains the number of individuals who can receive the later organizational opportunity. These are different forms of capacity operating at different stages of the process.

4.5.1. Selection capacity and scarcity.

Let N denote the eligible population entering the selection process and K the number of available opportunities. The ratio K/N defines the selection fraction, or equivalently the

degree of scarcity imposed by the selection process. Smaller values of K/N correspond to more restrictive selection, while larger values indicate that a greater proportion of the eligible population can receive the outcome.

The selection fraction matters because the consequences of a change in S_i depend on where that individual lies relative to the selection boundary. A score change that has no effect under relatively abundant selection may become consequential when only a small fraction of the population can be selected. Thus, congestion can alter competitive standing in SCORE without necessarily altering outcomes; whether that score change becomes consequential depends partly on the scarcity imposed in SELECT. In the deterministic case, the selection margin is located at the boundary separating the K highest scores from the remaining $N - K$ individuals.

4.5.2. Deterministic top- K selection

The simplest SELECT rule ranks individuals by S_i and awards the available opportunities to the K highest-scoring individuals. Under this deterministic top- K rule, every individual above the selection boundary is selected and every individual below it is not.

This formulation makes the distinction between SCORE and SELECT particularly transparent. SCORE determines the ordering of candidates; SELECT imposes a capacity constraint on that ordering. A congestion penalty can therefore change scores or even reorder some individuals without changing the final set of selected individuals if those changes occur away from the selection margin. Only rank changes that cross the top- K boundary alter the identities of the winners.

When exact score ties occur at the selection boundary, the implementation requires a deterministic tie-breaking rule. In the current experiments, exact ties are resolved using a fixed ordering so that repeated runs remain reproducible rather than introducing additional selection randomness. Random assignment in ASSIGN should not be confused with random selection in SELECT. The two processes introduce uncertainty at different stages and answer different scientific questions.

4.5.3. Stochastic selection and selection temperature.

Many selection processes contain uncertainty even after competitive scores have been established. A higher score can increase an individual's chance of selection without guaranteeing a place. To represent this uncertainty while preserving a fixed number of opportunities, we define a stochastic SELECT rule that draws exactly K distinct individuals without replacement. This is the intended fixed-capacity rule adopted for stochastic replay in the model's development history. Earlier deterministic experiments remain separate implementations of SELECT. [1]

Let $\mathcal{I}$ be the eligible population, with N individuals and an integer capacity satisfying $0 \leq K \leq N$. Each individual enters SELECT with the score $S_i = A_i - \lambda C_{g(i)}$ defined in Section 4.4.5. Scores and group membership remain fixed throughout selection. We denote the selection temperature by $t_{\text{SELECT}} > 0$, distinguishing it from the temperature parameter used in the earlier empirical fitting procedure, t_{MLE} , discussed in Section 4.8.

Selection proceeds through successive draws. If I_r denotes the individual chosen at draw r , the population remaining before that draw is

$$R_r = \mathcal{I} \setminus \{I_1, \dots, I_{r-1}\}, r = 1, \dots, K$$

For each $i \in R_r$, define the conditional probability of the next draw as

$$\Pr(I_r = i | I_1, \dots, I_{r-1}, \mathbf{S}) = \frac{\exp(S_i/t_{\text{SELECT}})}{\sum_{h \in R_r} \exp(S_h/t_{\text{SELECT}})}$$

Previously selected individuals have probability zero of being selected again. After each draw, the selected individual is removed and the denominator is recomputed over those who remain. The final outcome is therefore

$$Y_i = 1\{i \in \{I_1, \dots, I_K\}\}, \sum_{i \in \mathcal{I}} Y_i = K$$

These equations give an explicit statement of the sequential weighted sampling procedure described in the development records. The exponential terms are often called Gibbs weights, while sequential proportional sampling is also described there as a Plackett–Luce procedure. The displayed conditional probability describes the next draw. It is not the marginal probability that individual i belongs to the final set of K winners. The latter depends on all possible sequences through which that set can be formed. Draw order is a device for defining the selection distribution; it need not represent an observed institutional ranking. [1, 2]

Temperature controls how strongly score differences influence each draw. For two remaining individuals, the ratio of their next-draw probabilities is

$$\frac{\Pr(I_r = i | \cdot)}{\Pr(I_r = h | \cdot)} = \exp\left(\frac{S_i - S_h}{t_{\text{SELECT}}}\right)$$

Thus, lowering temperature magnifies the advantage associated with a given score difference. As temperature approaches zero, the selected set approaches the deterministic top- K set when there is no tie at the selection boundary. Boundary ties require a stated tie convention; the limiting stochastic rule does not itself impose the fixed-order convention used in deterministic code. As temperature becomes arbitrarily large, the next draw approaches uniform sampling from the remaining population and each individual's final inclusion probability approaches K/N . Temperature changes uncertainty in SELECT without changing the scores or the number of opportunities.

The mathematical rule permits any finite real score. Negative scores still yield positive exponential weights, and adding the same constant to every score leaves every draw probability unchanged. This matters when comparing the definition with legacy code. The shared selection helper first replaces negative scores by zero and sets the draw count to the smaller of K and the number of strictly positive scores. If no score is positive, it returns no selections. It also uses deterministic fallbacks at very small temperatures or when the numerical weights collapse. Consequently, some historical runs through that helper need not follow the unrestricted equation above or return exactly K winners. The cap does not mean that every zero-clipped candidate is excluded: when Gibbs sampling occurs, such candidates receive positive exponential weights. [3]

The equation above defines the intended model. Establishing which completed runs implement it requires identifying the code version and selection path used in each run, checking whether clipping, capping, or numerical fallbacks were active, and determining whether any reported result changes under an implementation that preserves the stated rule. That verification remains open. It should be completed before describing all legacy stochastic results as realizations of this exact model.

4.5.4. Fixed-capacity and Bernoulli selection are not equivalent

Stochastic selection can be implemented in more than one way depending on whether the organization fixes the number of available opportunities in advance. Earlier calibration work in this project used a Bernoulli formulation, in which each individual's selection was treated independently and the realized number selected could vary. That formulation is retained here as a model variant rather than the canonical fixed-capacity selection rule. By contrast, the fixed-capacity approach holds the total number of opportunities at K and selects exactly K individuals without replacement, with selection probabilities influenced by their scores.

These procedures can generate different outcome distributions even when they begin from the same scores. Bernoulli selection allows the realized number of winners to fluctuate, whereas exact- K selection preserves the selection capacity by construction. As a result, parameters calibrated under a Bernoulli process cannot be transferred automatically to an exact- K rule. Any such transfer requires explicit verification.

4.5.5. Mapping selection scarcity to empirical domains

The model treats K/N as a system-level parameter rather than a universal constant. Different organizational domains impose different levels of scarcity, and even within a domain the appropriate selection fraction depends on how the eligible population and outcome are defined. Consequently, empirical outcome rates cannot be inserted into the model without first matching the numerator and denominator to the modeled selection process.

For example, the approximately 2.7% value used in one basketball-related scenario reflected a particular empirical outcome proportion rather than a verified annual selection fraction for every simulated season. It should therefore be treated as a domain-motivated scenario value rather than a universal basketball selection rate.

SELECT therefore completes the model architecture. ASSIGN determines the peer environment an individual experiences, SCORE determines how that environment modifies competitive standing, and SELECT determines how many and which individuals ultimately receive the organizational distinction. Keeping these stages separate allows the model to ask whether a change in observed outcomes arises from network formation, environmental scoring, selection scarcity, or an interaction among them.

4.6. Parameter regimes: what each knob actually does

4.6.1. ρ : assignment preference and realized sorting

[FIGURE PLACEHOLDER: Holding all other parameters fixed and varying ρ across 4–5 values from random assignment to strong homophilic preference, I'll superimpose the resulting outcome curves and report the corresponding H_{sort} for each condition so that changes in generative preference can be compared with changes in realized sorting.]

4.6.2. γ : sharpness of the congestion transition

[FIGURE PLACEHOLDER: Holding all other parameters fixed and varying γ across 4–5 values, I'll superimpose the resulting outcome curves to show how a gradual versus sharp transition into competitive relevance changes the effect of congestion.]

4.6.3. θ : location of the competitive threshold

[FIGURE PLACEHOLDER: Holding all other parameters fixed and varying θ across 4–5 values spanning plausible regions of the A_i distribution, I'll superimpose the resulting outcome curves to show how changing which individuals count as competitively relevant shifts the congestion mechanism.]

4.6.4. λ : strength of the congestion penalty

[FIGURE PLACEHOLDER: Holding all other parameters fixed and varying λ across 4–5 values beginning at $\lambda = 0$, I'll superimpose the resulting outcome curves to show when congestion is too weak to alter rankings, when it begins to change outcomes, and how stronger penalties reshape the modeled relationship.]

4.6.5. K/N : scarcity and the selection boundary

[FIGURE PLACEHOLDER: Holding ASSIGN and SCORE parameters fixed and varying K/N across 4–5 levels of selection scarcity, I'll superimpose the resulting outcome curves to show how the same score distribution can produce different outcomes as the selection boundary becomes more or less restrictive.]

4.6.6. t : deterministic versus noisy selection

[FIGURE PLACEHOLDER: Holding ASSIGN, SCORE, and K fixed while varying selection temperature t across 4–5 values, I'll superimpose the resulting outcome curves to show how increasing selection noise changes the translation from competitive score into observed outcomes.]

4.6.7. Interactions among parameters: where similar curves can come from different mechanisms

[FIGURE PLACEHOLDER: After presenting the one-parameter sweeps above, I'll select one theoretically important two-parameter interaction and show a bounded comparison illustrating that similar outcome curves can arise from different combinations of mechanisms. I'll avoid expanding into a full multidimensional parameter search.]

4.7. Mechanism tests and model behavior

- 4.7.1. Can congestion change winners without similarity preference? Separate this question from whether a downturn occurs.
- 4.7.2. Was the initial penalty large enough to reshape the curve? Explain the penalty's scale relative to individual performance.
- 4.7.3. How does curve shape depend on penalty strength and binning? Show equal-width and quantile views with support counts.
- 4.7.4. Does a low selection fraction make assignment preference irrelevant? Distinguish the number of changed winners from the fraction of winners changed.
- 4.7.5. How does reduced preference alter sorting, the range of peer environments, and selection profiles? Treat empty bins as missing support.
- 4.7.6. Robustness across model versions and experimental choices. Use the existing plot trail, retaining model versions, dates, and the purpose of the 50% selection comparison. Distinguish exploratory choices from estimated optima.

4.8. Model scope, limitations, and the bridge to empirical fitting

The empirical fitting work estimated parameters on fixed observed basketball rosters using a probability model distinct from the fixed-capacity SELECT rule. The fitting procedure did not regenerate teams through ASSIGN or simulate a set of winners during optimization. Instead, it evaluated how well a season-wise probability specification accounted for the recorded binary outcomes on the observed rosters. The existing estimates therefore document a completed Bernoulli calibration. They do not constitute an exact- K likelihood fit of the stochastic model defined in Section 4.5.3. [4]

For season s , let $\mathcal{J}_s$ denote the player-season observations entering the likelihood. The measured performance quantity A_{is} was points per minute standardized within season by the analysis pipeline. The binary outcome Y_{is} indicated whether the player was ever drafted. The fitting panel retained all eligible player-season observations rather than restricting outcomes to one final-season observation per athlete. Accordingly, the same eventual draft outcome could appear on several rows belonging to one player. These are the observational units of the historical fit; they should not be described as independent annual draft decisions or as distinct draft selections merely because the likelihood multiplies contributions across rows. A revision intended to model annual selection would need to establish a corresponding eligible population and outcome timing. [5]

Congestion was computed on the observed team-season rosters using the full-group definition introduced in Section 4.4.5. With $g_s(i)$ denoting the group containing individual i in season s , the fitted congestion term was

$$C_{js}(\gamma, \theta_s) = \frac{1}{n_{js}} \sum_{h: g_s(h)=j} \sigma(\gamma(A_{hs} - \theta_s)), \sigma(x) = \frac{1}{1 + \exp(-x)}$$

The mean included the focal player as well as the other retained members of the group. In the code this quantity was stored as `pool_c_smooth_team` and was often denoted L^C in development documents. Here we use C_{js} to preserve the chapter's notation. This congestion quantity differs from the leave-one-out peer-quality mean used on some HERO horizontal axes. A plot

conditioned on peer quality and a likelihood using congestion can coexist, but the two quantities must remain identified separately when interpreting an empirical comparison. [6]

The intermediate quantity supplied to the probability calculation, called the board logit in the implementation, was

$$\eta_{is} = \frac{A_{is}}{t_{MLE}} - \lambda_{MLE} C_{g_s(i),s}(\gamma, \theta_s)$$

The script converted these quantities to probabilities by applying a softmax function within each season:

$$p_{is} = \frac{\exp(\eta_{is})}{\sum_{h \in \mathcal{I}_s} \exp(\eta_{hs})}, \sum_{i \in \mathcal{I}_s} p_{is} = 1$$

The softmax specifies the probabilities; the Bernoulli model specifies how those probabilities contribute to the likelihood of the observed binary outcomes. The historical likelihood and log-likelihood were

$$\begin{aligned} \mathcal{L}_B(\lambda_{MLE}, \gamma, t_{MLE}) &= \prod_s \prod_{i \in \mathcal{I}_s} p_{is}^{Y_{is}} (1 - p_{is})^{1-Y_{is}}, \\ \ell_B(\lambda_{MLE}, \gamma, t_{MLE}) &= \sum_s \sum_{i \in \mathcal{I}_s} [Y_{is} \log p_{is} + (1 - Y_{is}) \log(1 - p_{is})]. \end{aligned}$$

For numerical evaluation, the implementation clipped each probability to the interval $[10^{-15}, 1 - 10^{-15}]$ before taking logarithms. It minimized the negative log-likelihood using L-BFGS-B over the logarithms of the positive parameters λ_{MLE} , γ , and t_{MLE} . Congestion was recomputed as γ changed during joint optimization. A separate option held γ fixed and optimized the other two parameters. Reported overlap between the highest-probability observations and positive outcomes was a diagnostic, not the objective being optimized. [4]

The season normalization has a consequential implication. Before numerical clipping, the working independent Bernoulli model, conditional on the thresholds treated as fixed during fitting, satisfies

$$E \left[\sum_{i \in \mathcal{I}_s} Y_{is} \mid \mathbf{A}, \mathbf{g}, \theta_s \right] = \sum_{i \in \mathcal{I}_s} p_{is} = 1$$

The code did not multiply the probabilities by the observed number of positive outcomes, fit a season intercept to that count, or condition the likelihood on selecting exactly K_s individuals. This is the historical working Bernoulli specification, but its normalization does not match a season containing many observed positive outcomes. Consequently, the saved probabilities should not be presented as validated individual draft probabilities. What remains to be established is whether this objective is defensible as a working calibration criterion for the intended scientific comparison, or whether a different outcome likelihood and observational unit are required. Neither a satisfactory ranking diagnostic nor a successful optimizer convergence message resolves that question.

The viability threshold θ_s was fixed during optimization, although its value was calculated separately for each season. The preprocessing routine counted positive draft labels among that season's rows with observed performance, used $K_s^Y = \max(1, \sum_i Y_{is})$, and set

$$\theta_s = Q_{1-K_s^Y/N_s}(\{A_{is}\})$$

where N_s and the empirical quantile refer to that preprocessing population. The superscript Y distinguishes this count of positive labels from a separately established institutional capacity. Because the labels describe eventual drafting on player-season rows, their season total is not automatically the number drafted in that calendar year. Thus, the outcome count affected the fit indirectly through the construction of congestion, even though the Bernoulli likelihood imposed no fixed-capacity constraint. The interpretation and timing of this threshold must be revisited if the outcome population is changed. [7]

Assignment preference ρ was outside the outcome likelihood because team membership was observed and held fixed. A separate calibration compared simulated and empirical values of the sorting statistic H_{sort} . The saved longitudinal result selected $\rho = 0$, at the lower boundary of the assignment calibration. The empirical mean sorting statistic remained below the simulated mean at that boundary, so this result does not establish that actual team assignment was random. It describes the closest setting reported for that calibration. The Gibbs selection temperature was also treated separately: it was swept in simulation, with a replay starting value of one, rather than jointly estimated as part of an exact- K likelihood. [1, 8]

The joint fitting artifact saved on 28 August 2026 reports

$$\hat{\gamma} = 19.57233208, \hat{\lambda}_{\text{MLE}} = 1.30243058, \hat{t}_{\text{MLE}} = 1.06989673,$$

with log-likelihood -8865.70116079, 46,306 player-season observations, and 1,133 positive outcome flags across 11 season batches. The optimizer reported convergence. These are historical point estimates, not evidence by themselves of precise identification, causal effects, or a validated generative selection model. [9]

The run's analysis-window documentation is internally inconsistent. Its filename, saved metadata, and campaign description identify 2009–2021. However, the inspected panel loader constructs a 2011–2021 panel before applying the requested window, which is consistent with the saved total of 11 seasons. We therefore report both the recorded 2009–2021 run label and the implemented 2011–2021 restriction rather than asserting that the fit contains all thirteen seasons. Confirmation of the original run's input seasons and code version remains necessary to finalize its sample description. An earlier 2013–2021 fit, with $\gamma = 18$ fixed, instead reported $\hat{\lambda}_{\text{MLE}} = 2.57150065$ and $\hat{t}_{\text{MLE}} = 1.06556159$. Those estimates belong to a different fitting specification and should not be merged with the later joint estimates. [5, 9, 10]

There is a further distinction between the scale of the fitted logit and the score used in the generative model. In the fitting code, temperature divides measured performance alone; in stochastic SELECT it divides the entire score. Algebraically,

$$\frac{A_i}{t_{\text{MLE}}} - \lambda_{\text{MLE}} C_{g(i)} = \frac{A_i - t_{\text{MLE}} \lambda_{\text{MLE}} C_{g(i)}}{t_{\text{MLE}}}$$

For fixed congestion values, reproducing those same logits with a score of the chapter's form would require a score coefficient $\lambda = t_{\text{MLE}} \lambda_{\text{MLE}}$ and, for the Gibbs weights, $t_{\text{SELECT}} = t_{\text{MLE}}$. This identity clarifies the change of parameter scale; it is not a historical decision to equate the two temperatures or evidence that their likelihoods are equivalent. Even ranking by the fitted probabilities need not agree with ranking by $A_i - \lambda_{\text{MLE}} C_{g(i)}$ when the numerical penalty coefficient is transferred unchanged. [4, 11]

An exact- K fit would evaluate the probability of the observed selected set under the without-replacement process. If selection order is unobserved, that probability sums over the possible orders of the selected individuals. It is not the product of the independent Bernoulli contributions used above. Development documents considered such a likelihood, but explicitly retained Bernoulli estimation for the completed fitting procedure. No completed exact- K refit has been established in the verified record. [2, 12]

The empirical bridge consequently remains a separate methodological task. Before treating the historical estimates as fitted parameters of deterministic top- K or stochastic exact- K selection, the analysis must align the outcome population and timing, the congestion definition and scale, the selection implementation, and the probability model used for estimation. A refit or an explicit justification for using the earlier estimates is still required. This distinction allows the completed fitting work and mechanism experiments to be documented together while preserving the limits of what their combination currently establishes. [13]